

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 08

DATE : 04-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
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REVISION TEST # 08

Topics

DATE : 04 - 12 - 2019**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS** : Rotational Mechanics**CHEMISTRY** : Alkyl Halide, Oxidation + Reduction + Grignard Reagent**BOTANY** : Plant Diversity (Fungi, Algae), Nucleic Acids, Central Dogma

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. A uniform circular disc A of radius r is made from a copper plate of thickness t and another uniform circular disc B of radius $2r$ is made from a copper plate of thickness $t/2$. The relation between the moments of inertia I_A and I_B is

- (1) $I_A > I_B$
(2) $I_A = I_B$
(3) $I_A < I_B$
(4) depends on the values of t and r .

2. The instantaneous angular position of a point on a rotating wheel is given by the equation :-

$$\theta = 2t^3 - 6t^2$$

The torque on the wheel becomes zero at

- (1) $t = 0.5$ sec (2) $t = 0.25$ sec
(3) $t = 2$ sec (4) $t = 1$ sec

3. The grinding stone of a flour mill is rotating at 600 rad/sec. for this power of 1.2 k watt is used. the effective torque on stone in N-m will be :-

- (1) 1 (2) 2
(3) 3 (4) 4

4. The radius of gyration of a rod of length L and mass M about an axis perpendicular to its length and passing through a point at a distance $L/3$ from one of its ends is :-

- (1) $\frac{\sqrt{7}}{6}L$ (2) $\frac{L}{9}$
(3) $\frac{L}{3}$ (4) $\frac{\sqrt{5}}{2}L$

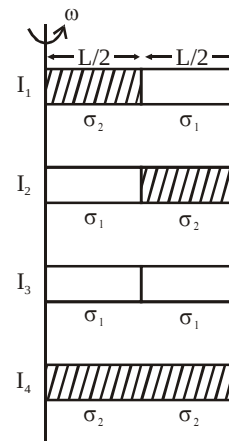
5. A constant power is supplied to a rotating disc. Angular velocity (ω) of disc varies with number of rotations (n) made by the disc as:

- (1) $\omega \propto (n)^{1/3}$ (2) $\omega \propto (n)^{3/2}$
(3) $\omega \propto (n)^{2/3}$ (4) $\omega \propto (n)^2$

6. We have two spheres one of which is hollow and the other solid. They have identical masses and moment of inertia about their respective diameters. The ratio of their radius is given by:-

- (1) 5 : 7 (2) 3 : 5
(3) $\sqrt{3} : \sqrt{5}$ (4) $\sqrt{3} : \sqrt{7}$

- 7.

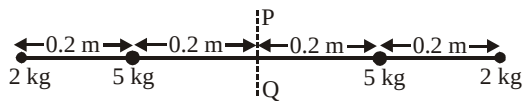


If all rods have same length and area of crosssection. Also density (σ_2) > density (σ_1) then :-

- (1) $I_4 > I_2 > I_3 > I_1$ (2) $I_4 > I_3 > I_2 > I_1$
(3) $I_1 > I_2 > I_3 > I_4$ (4) $I_4 > I_2 > I_1 > I_3$

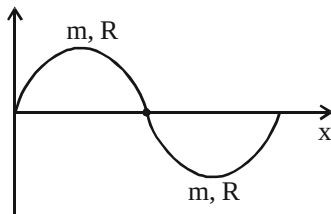
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8. Four masses are fixed on a massless rod as shown in the figure. The moment of inertia about the axis PQ is about :-



- (1) 2 kg-m^2 (2) 1.04 kg-m^2
(3) 0.5 kg-m^2 (4) 0.3 kg-m^2

9. There are two semi-circular rings each of mass m and radius R , arranged so as to form a shape similar to a sine curve. Calculate its moment of inertia about the x -axis :-



- (1) mR^2 (2) $\frac{mR^2}{2}$
(3) $2mR^2$ (4) None of these

10. The moment of inertia of a solid sphere about an axis passing through centre of gravity is $\frac{2}{5} MR^2$; then its radius of gyration about a parallel axis at a distance $2R$ from first axis is:-

- (1) $5R$ (2) $\sqrt{22/5}R$
(3) $\frac{5}{2}R$ (4) $\sqrt{12/5}R$

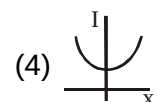
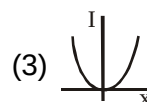
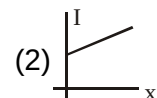
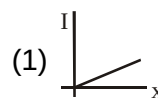
11. A circular disc X of radius R is made from an iron plate of thickness t and another disc Y of radius $4R$ is made from an iron plate of thickness $t/4$. Then the relation between the moment of inertia I_x and I_y is :-

- (1) $I_y = 32 I_x$
(2) $I_y = 16 I_x$
(3) $I_y = I_x$
(4) $I_y = 64 I_x$

12. Axis X and Z lying in the plane of the disc are mutually perpendicular and axis Y is perpendicular to the plane of the disc. If the moment of inertia of the disc about the axes X and Y are 20 kg-m^2 and 40 kg-m^2 respectively then the moment of inertia of the disc about Z-axis would be

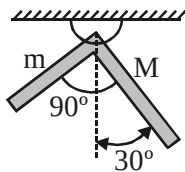
- (1) 60 kg-m^2
(2) 40 kg-m^2
(3) 20 kg-m^2
(4) Zero

13. Figure represents the moment of inertia of the solid sphere about an axis parallel to the diameter of the solid sphere and at a distance x from it. Which one of the following represents the variation of I with x :-



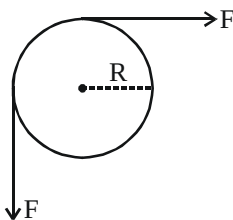
SPACE FOR ROUGH WORK

14. Two uniform rods of equal length but different masses are rigidly joined to form an L-shaped body, which is then pivoted as shown in figure. If in equilibrium the body is in the shown configuration, ratio M/m will be :-



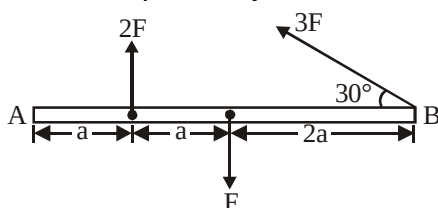
- (1) 2 (2) 3
(3) $\sqrt{2}$ (4) $\sqrt{3}$

15. Two equal forces are applied tangentially to a uniform disc of mass M and radius R as shown in the figure. If the disc is pivoted at its centre and free to rotate in its plane, the angular acceleration of the disc is :-



- (1) $\frac{F}{MR}$ (2) $\frac{\sqrt{2}F}{MR}$ (3) $\frac{2F}{MR}$ (4) Zero

16. Three forces F , $2F$ and $3F$ act on a rod AB which is pivoted at A . The anticlockwise moment of forces F , $2F$ and $3F$ about the pivot are respectively :-

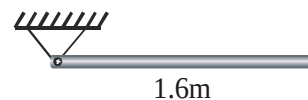


- (1) $+2Fa; -3Fa; +6Fa$
(2) $-2Fa; +2Fa; +6Fa$
(3) $-2Fa; -2Fa; -6Fa$
(4) $+2Fa; -2Fa; +6Fa$

17. What is the torque of force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ acting at a point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about the origin?

- (1) $6\hat{i} - 6\hat{j} + 12\hat{k}$ (2) $-6\hat{i} + 6\hat{j} - 12\hat{k}$
(3) $17\hat{i} - 6\hat{j} - 13\hat{k}$ (4) $-17\hat{i} + 6\hat{j} + 13\hat{k}$

18. The uniform rod of mass 20 kg and length 1.6 m is pivoted at its end and swings freely in the vertical plane. Angular acceleration of rod just after the rod is released from rest in the horizontal position as shown in figure is



- (1) $\frac{15g}{16}$ (2) $\frac{17g}{16}$ (3) $\frac{16g}{15}$ (4) $\frac{g}{15}$

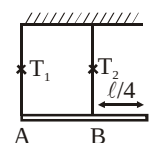
19. A uniform rod of length l and mass M is hung from two strings of equal length from a ceiling as shown. Tension in the two strings is :-

(1) $T_1 = \frac{Mg}{3}, T_2 = \frac{2Mg}{3}$

(2) $T_1 = T_2 = \frac{Mg}{2}$

(3) $T_1 = \frac{2Mg}{3}, T_2 = \frac{Mg}{3}$

(4) $T_1 = Mg, T_2 = 0$



SPACE FOR ROUGH WORK

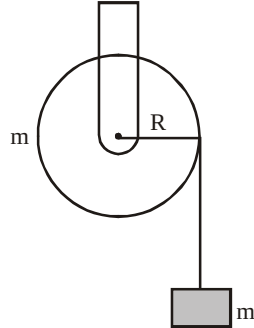
20. A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?

(1) $\frac{5g}{6}$

(2) g

(3) $\frac{2g}{3}$

(4) $\frac{g}{2}$



21. A cubical block of mass M and edge a slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about its centre has a magnitude. :-

(1) zero

(2) Mga

(3) $Mg \sin \theta$

(4) $\frac{Mg \sin \theta}{2}$

22. A particle is rotating along a circular path in the X-Y plane. The angular momentum vector of the particle will be directed parallel to :-

(1) x-axis

(2) y-axis

(3) z-axis

(4) none of these

23. A uniform metre stick is held vertically with one end on the floor and is allowed to fall. The speed of the other end when it hits the floor assuming that the end at the floor does not slip:

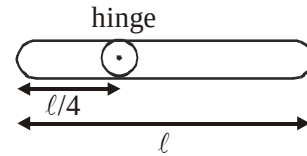
(1) $\sqrt{4g}$

(2) $\sqrt{3g}$

(3) $\sqrt{5g}$

(4) $\sqrt{g}$

24. A uniform rod is hinged as shown in the figure and is released from a horizontal position. The angular velocity of the rod as it passes the vertical position is:



(1) $\sqrt{\frac{12g}{3l}}$

(2) $\sqrt{\frac{2g}{3l}}$

(3) $\sqrt{\frac{24g}{7l}}$

(4) $\sqrt{\frac{3g}{7l}}$

25. A solid sphere of mass m and radius r rolls on a horizontal plane without slipping with the speed u. Now, if it rolls up vertically, the maximum height it would attain will be :-

(1) $3u^2/4g$

(2) $5u^2/2g$

(3) $7u^2/10g$

(4) $u^2/2g$

26. A solid sphere, a hollow sphere and a ring, all having equal mass and radius, are placed at the top of an incline and released. The friction coefficients between the objects and the incline are equal and but not sufficient to allow pure rolling. The greatest kinetic energy at the bottom of the incline will be achieved by

(1) the solid sphere

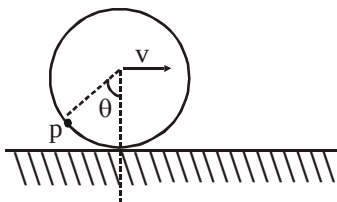
(2) the hollow sphere

(3) the ring

(4) all will achieve same kinetic energy.

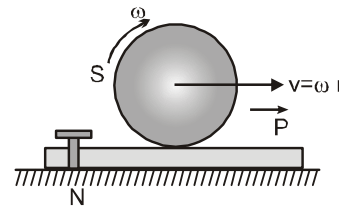
SPACE FOR ROUGH WORK

27. When a solid sphere of moment of inertia I rolls down on an inclined plane the percentage of total energy which is rotational is approximately :-
 (1) 28% (2) 72% (3) 100% (4) None
28. A solid sphere rolls down on an inclined plane of inclination θ . What is the acceleration as the sphere reaches bottom:-
 (1) $\frac{5}{7} g \sin \theta$ (2) $\frac{3}{5} g \sin \theta$
 (3) $\frac{2}{7} g \sin \theta$ (4) $\frac{2}{5} g \sin \theta$
29. A spherical uniform body of radius R , mass M and moment of inertia I rolls down (without slipping) on an inclined plane making an angle θ with the horizontal. Then its acceleration is:-
 (1) $\frac{g \sin \theta}{1 - MR^2/I}$ (2) $\frac{g \sin \theta}{1 + I/MR^2}$
 (3) $\frac{g \sin \theta}{1 + MR^2/I}$ (4) $\frac{g \sin \theta}{1 - I/MR^2}$
30. A hoop rolls on a horizontal ground without slipping with linear speed v . Speed of a particle P on the circumference of the hoop at angle θ as shown is :



- (1) $2v \sin \frac{\theta}{2}$ (2) $v \sin \theta$
 (3) $2v \cos \frac{\theta}{2}$ (4) $v \cos \theta$

31. A sphere S rolls without slipping, moving with a constant speed on a plank P . The friction between the upper surface of P and the sphere is sufficient to prevent slipping, while the lower surface of P is smooth and rests on the ground. Initially, P is fixed to the ground by a pin N . If N is suddenly removed:



- (1) S will begin to slip on P
 (2) P will begin to move backwards
 (3) the speed of S will decrease and its angular velocity will increase
 (4) there will be no change in the motion of S and P will still be at rest.
32. Two bodies have their moments of Inertia I and $2I$ respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momentum will be in the ratio :-
 (1) 1:2 (2) $\sqrt{2}:1$
 (3) $1:\sqrt{2}$ (4) 2:1
33. A ball is thrown on a lawn in such a way that it initially slides with a speed v_0 without rolling. It gradually picks up rotation motion. Find the speed of the ball at which there will be rolling without slipping-

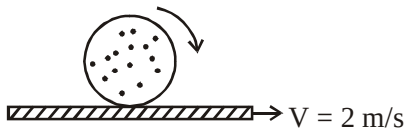
- (1) $\frac{2}{7} v_0$ (2) $\frac{2}{5} v_0$ (3) $\frac{5}{7} v_0$ (4) $\frac{3}{5} v_0$

SPACE FOR ROUGH WORK

34. A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is :-

- (1) $40\sqrt{\frac{5}{7}}$ m/s (2) 20 m/s
(3) 10 m/s (4) $10\sqrt{30}$ m/s

35. A uniform disc of radius 20 cm is performing pure rolling on a moving surface, if its angular velocity is 5 rad/s then find velocity of its centre of mass :-



- (1) 1 m/s (2) 2 m/s
(3) 3 m/s (4) None of these

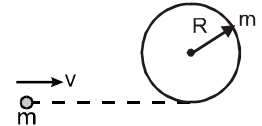
36. A rotating table completes one rotation in 10 sec and its moment of inertia is 100 kg-m^2 . A person of 50 kg. mass stands at the centre of the rotating table. If the person moves 2m. from the centre, the angular velocity of the rotating table in rad/sec. will be :-

- (1) $\frac{2\pi}{30}$ (2) $\frac{20\pi}{30}$ (3) $\frac{2\pi}{3}$ (4) 2π

37. A uniform thin circular ring of mass 'M' and radius 'R' is rotating about its fixed axis with a constant angular velocity ω . Two objects each of mass m, are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with an angular velocity.

- (1) $\frac{\omega M}{(M+m)}$ (2) $\frac{\omega M}{(M+2m)}$
(3) $\frac{\omega M}{(M-2m)}$ (4) $\frac{\omega(M+3m)}{M}$

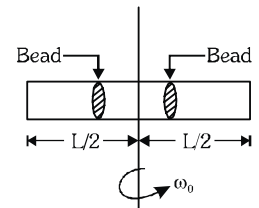
38. A circular wooden loop of mass m and radius R rests flat on a horizontal frictionless surface.



A bullet, also of mass m, and moving with a velocity V, strikes the loop and gets embedded in it. The thickness of the loop is much smaller than R. The angular velocity with which the system rotates just after the bullet strikes the loop is

- (1) $\frac{V}{4R}$ (2) $\frac{V}{3R}$ (3) $\frac{2V}{3R}$ (4) $\frac{3V}{4R}$

39. A smooth uniform rod of length L and mass M has two identical beads of negligible size, each of mass m, which can slide freely along the rod.



Initially the two beads are at the centre of the rod and the system is rotating with angular velocity ω_0 about its axis perpendicular to the rod and passing through its mid-point (see figure). There are no external forces. When the beads reach the ends of the rod, the angular velocity of the system is:

- (1) $\frac{M\omega_0}{M+3m}$ (2) $\frac{M\omega_0}{M+6m}$
(3) $\frac{(M+6m)\omega_0}{M}$ (4) ω_0

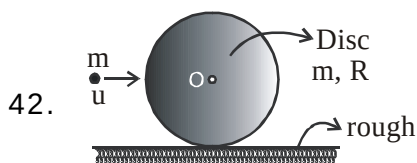
SPACE FOR ROUGH WORK

40. A chain couples and rotates two wheels in a bicycle. The radii of bigger and smaller wheels are 0.5m. and 0.1m respectively. The bigger wheel rotates at the rate of 200 rotations per minute, then the rate of rotation of smaller wheel will be :-

(1) 1000 rpm (2) $\frac{50}{3}$ rpm
(3) 200 rpm (4) 40 rpm

41. A uniform rod of mass m length ℓ rests on a smooth horizontal surface. Rod is given sharp horizontal impulse p perpendicular to the rod at a distance $\ell/4$ from the centre. The angular velocity of rod will be-

(1) $\frac{3p}{m\ell}$ (2) $\frac{p}{m\ell}$ (3) $\frac{p}{2m\ell}$ (4) $\frac{2p}{m\ell}$

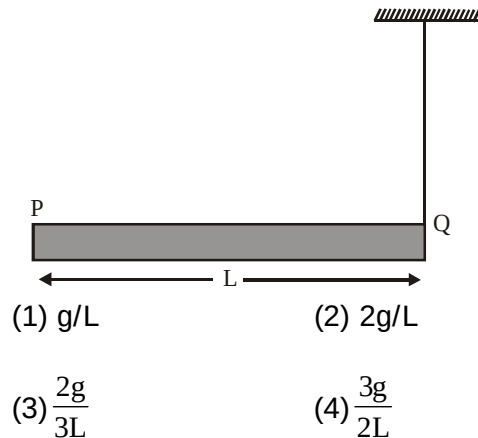


42.

A point mass m collides with a disc of mass m and radius R resting on a rough horizontal surface as shown in figure. Its collision is perfectly inelastic and immediately system starts pure rolling. Find angular velocity of the system after pure rolling starts :

(1) $\left(\frac{2u}{7R}\right)$ (2) $\left(\frac{7u}{2R}\right)$ (3) $\left(\frac{5u}{2R}\right)$ (4) $\left(\frac{2u}{5R}\right)$

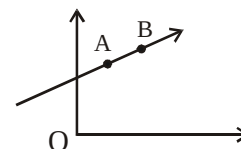
43. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is :-



44. A uniform disc of mass M and radius R is mounted on an axle supported in frictionless bearings. A light cord is wrapped around the rim of the disc and a steady downward pull T is exerted on the cord. The angular acceleration of the disc is :-

(1) $\frac{T}{MR}$ (2) $\frac{MR}{T}$ (3) $\frac{2T}{MR}$ (4) $\frac{MR}{2T}$

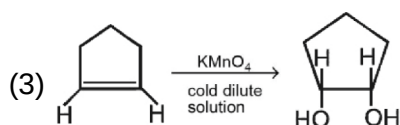
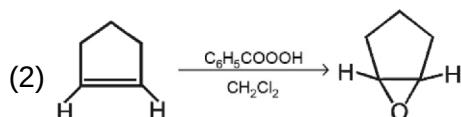
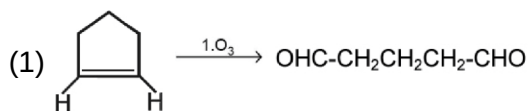
45. A particle of mass m moves in the X-Y plane with a velocity v along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at B, then:



(1) $L_A = L_B$
(2) $L_A > L_B$
(3) $L_A < L_B$
(4) none of these

SPACE FOR ROUGH WORK

46. Which of the following are correct ?



(4) All of these

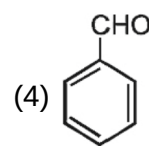
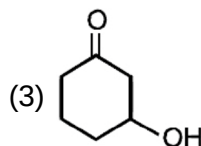
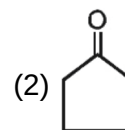
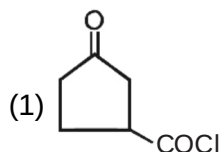
47. Which functional group can not be reduced by H_2/Pd at room temperature?



48. Clemmensen reduction is not preferred for compounds which have _____ group.

- (1) Base sensitive
- (2) Acid sensitive
- (3) both of the above
- (4) none of these

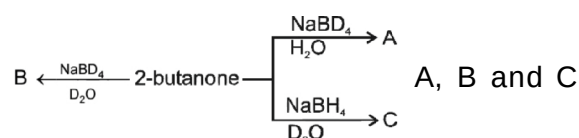
49. Wolff kishner reduction is not prefer in which of the following?



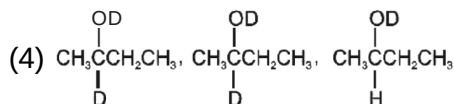
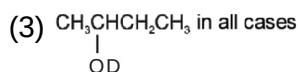
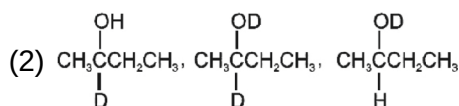
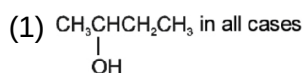
50. On oxidation, an unknown alkene C_8H_{16} gives a mixture of propanoic acid and pentanoic acid. The unknown alkene can be

- (1) cis-3-octene
- (2) trans-2-octene
- (3) cis-4-octene
- (4) trans-4-octene

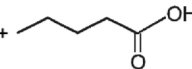
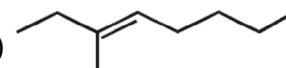
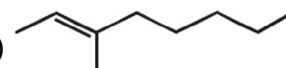
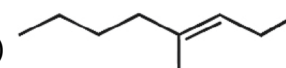
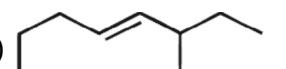
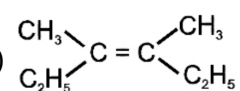
51. Consider reduction of 2-butanone.



are respectively.



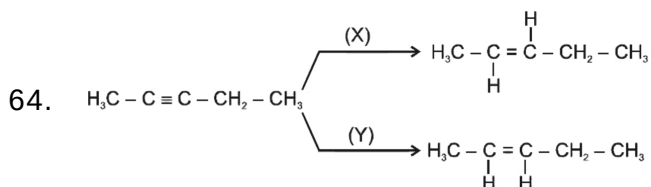
SPACE FOR ROUGH WORK

52. Acetone will be obtained on ozonolysis of
 (1) 1-pentene (2) 2-pentene
 (3) isopentene (4) 2-pentyne
53. Oxidation of cyclopentanol to cyclopentanone can be accomplished by using
 (1) Tollen's reagent
 (2) chromic acid
 (3) bromine water
 (4) Fehling's solution.
54. $\begin{array}{c} \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array} \xrightarrow{\text{HIO}_4}$ product obtained in the above reaction
 (1) 2HCHO (2) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$
 (3) 2HCOOH (4) $\begin{array}{c} \text{CHO} \\ | \\ \text{COOH} \end{array}$
55. Which of the following do not give glyoxal on ozonolysis followed by reaction with Zn and H₂O?
 (1) Benzene (2) HC $\equiv$ CH
 (3) Toluene (4) Ethene
56. $\text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow{\text{Reagent}} \text{CH}_3\text{CHO}$, Reagent is :-
 (1) KMnO₄ hot (2) K₂Cr₂O₇
 (3) PCC (4) CrO₃/H⁺
57. $\text{X} \xrightarrow[\text{O}_3]{\text{Ozonolysis}} \text{CH}_3\text{CHO} + \text{CH}_3\text{CH}_2\text{CHO}$, is :-
 (1) 2-butene (2) 2-pentene
 (3) 1-pentene (4) Iso-pentene
58. The complete combustion of CH₄ gives :
 (1) CO + H₂ (2) CO + N₂
 (3) CO + H₂O (4) CO₂ + 2H₂O
59. Alkene (X) $\xrightarrow[\Delta]{\text{KMnO}_4}$  + 
 X is :
 (1) 
 (2) 
 (3) 
 (4) 
60. Baeyer's reagent is :-
 (1) alkaline KMnO₄ solution
 (2) acidic KMnO₄ solution
 (3) neutral KMnO₄ solution
 (4) aqueous bromine solution
61. Reactant $\xrightarrow{\text{OsO}_4 / \text{H}_2\text{O}_2}$ Diketone, reactant is :-
 (1) Alkyne (2) Alkene
 (3) Alkane (4) Alcohol
62. An alkene with molecular formula C₈H₁₆ on oxidation with hot KMnO₄ gives acetone and 3-pentanone. The structure of the alkene is
 (1) (CH₃)₂C = C(C₂H₅)₂
 (2) 
 (3) (C₂H₅)₂OCHCH₂CH₃
 (4) (CH₃)₂C = CH(CH₂)₃CH₃.

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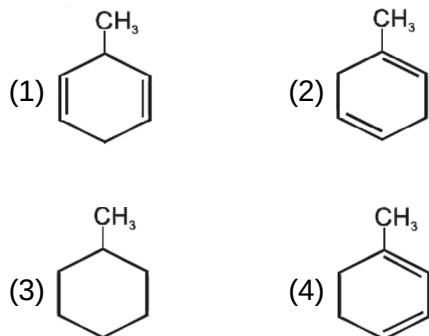
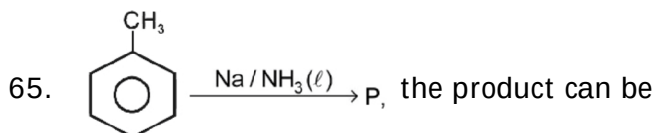
63. Jones's reagent is

- (1) Acidified $K_2Cr_2O_7$ solution
- (2) Alkaline $K_2Cr_2O_7$ solution
- (3) CrO_3 in aqueous acetone
- (4) A solution of $K_2Cr_2O_7/H_2SO_4$ in aqueous ethanol



In the above reaction the using reagents X and Y are :-

- (1) $Na/ liq. NH_3$ for X
- (2) $H_2, Pd/BaSO_4$ for Y
- (3) $BH_3-THF + CH_3COOH$ for Y
- (4) All of these



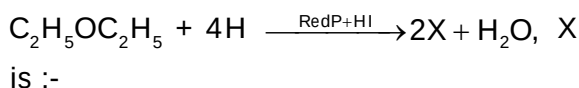
66. Ester $\xrightarrow{DIBAL-H/Cold}$ Product

- (1) Alcohol
- (2) Aldehyde
- (3) Ketone
- (4) Acid

67. An isocyanide on reduction with hydrogen in the presence of platinum gives :-

- (1) amide
- (2) primary amine
- (3) secondary amine
- (4) alcohol

68. In the following reaction



- (1) Ethane
- (2) Ethylene
- (3) Butane
- (4) Propane

69. For the conversion of CH_3OH into methane, the reagent used is :-

- (1) sodium
- (2) P and HI
- (3) hydrogen
- (4) $LiAlH_4$

70. Stephen reduction converts cyanides into

- (1) Aldehydes
- (2) Ketones
- (3) Amines
- (4) Acids

71. Which of the following is not organometallic compound

- (1) Grignard reagent
- (2) Gilman's reagent
- (3) Frankland reagent
- (4) Tilden reagent

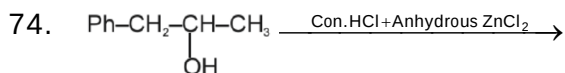
72. The reagent used to convert $RCOOH \longrightarrow RCH_2OH$ is

- (1) $NaBH_4$
- (2) $Na/Alcohol$
- (3) $Zn/Hg-HCl$
- (4) $LiAlH_4$

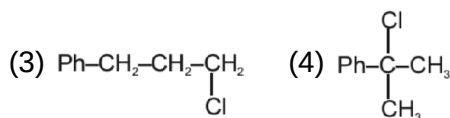
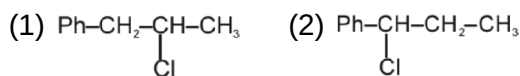
73. When benzoic anhydride is treated with $LiAlH_4$, it forms

- (1) Benzaldehyde
- (2) Benzyl alcohol
- (3) Benzene
- (4) Toluene

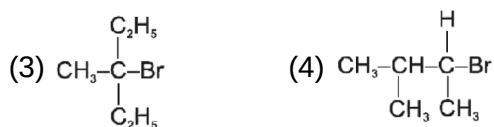
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X (Major product) X is :-



75. Which one of the following compound will give (d) and (l) form in $\text{S}_\text{N}1$ reaction (as major product)



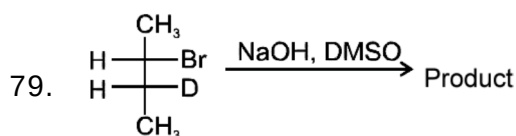
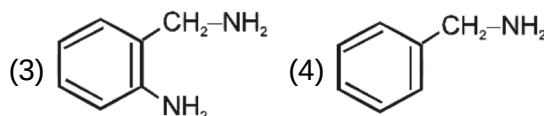
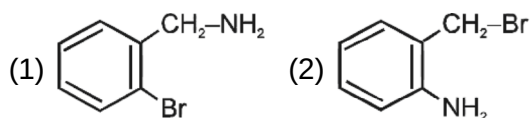
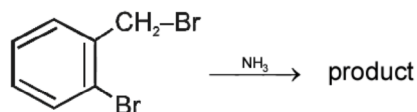
76. $\text{S}_\text{N}2$ mechanism proceeds through intervention of :-

- (1) Carbonium ion
- (2) Transition state
- (3) Free radical
- (4) Carbanion

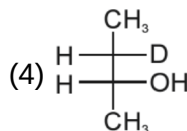
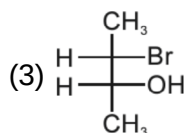
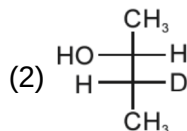
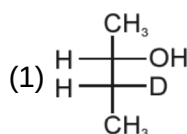
77. In $\text{S}_\text{N}2$ reaction if we doubled the concentration of reactant and nucleophile the rate of $\text{S}_\text{N}2$ reaction increases by :

- (1) 2 times
- (2) 4 times
- (3) 8 times
- (4) No change

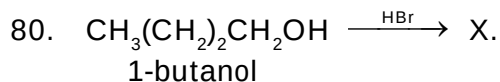
78. What is the major product obtained in the following reaction ?



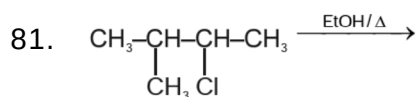
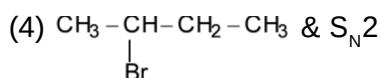
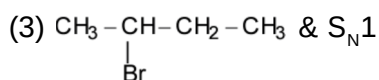
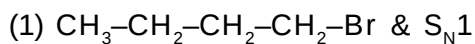
Product is :-



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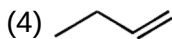
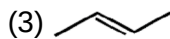
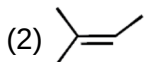
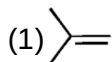
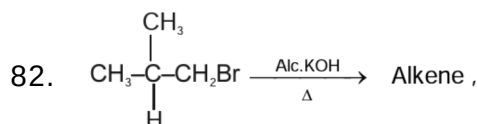
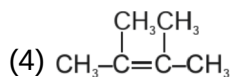
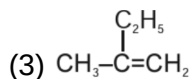
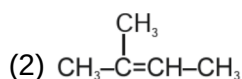
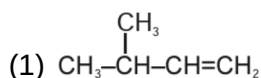


Identify X and the type of mechanism of the reaction?

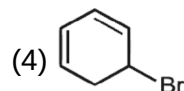
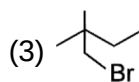
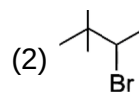
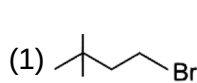


Major elimination product X.

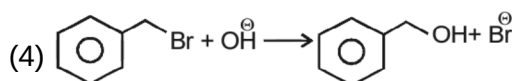
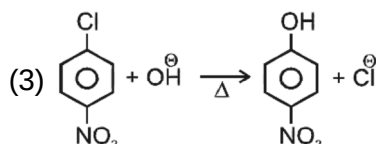
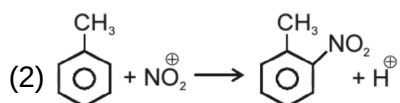
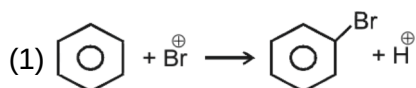
X is :-



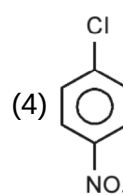
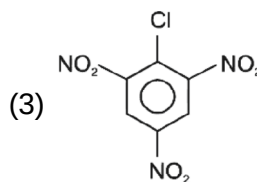
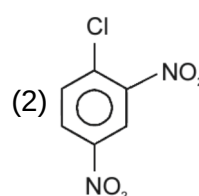
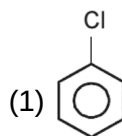
83. Which of the following cannot undergo E_2 reaction ?



84. Which of the following reaction is $\text{S}_{\text{N}}2\text{Ar}$ reaction ?

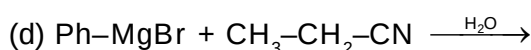
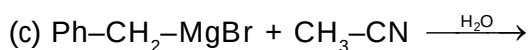
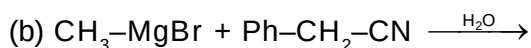
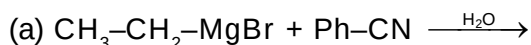


85. Which of the following compound gives fastest $\text{S}_{\text{N}}2\text{Ar}$ reaction?



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86. Which of the following can give 1-phenyl propan-2-one ?



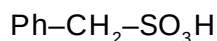
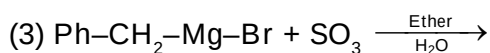
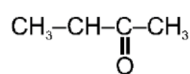
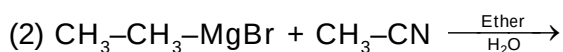
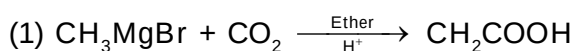
(1) (a) and (b)

(2) (b) and (c)

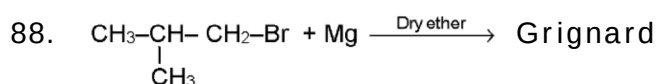
(3) (c) and (d)

(4) (a) and (d)

87. Which of the following reactions are nucleophilic addition reaction ?

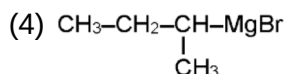
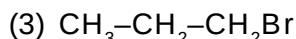
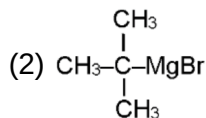
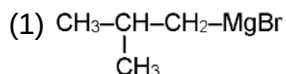


(4) All of these

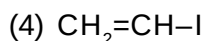
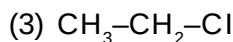
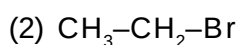
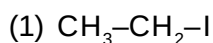


Reagent

Grignard reagent is :-



89. Which of the following alkyl halide is most reactive towards Grignard reagent formation with Mg in presence of dry ether.



90. Grignard reagent can give :-

(1) Nucleophilic substitution reaction

(2) Nucleophilic addition reaction

(3) Acid-Base reaction

(4) All of these

SPACE FOR ROUGH WORK

91. In oogamy, fertilization involves :
- (1) A large motile female gamete and a small non-motile male gamete
 - (2) A small non-motile female gamete and a large motile male gamete
 - (3) A large non-motile female gamete and a small motile male gamete
 - (4) A large motile female gamete and a large non-motile male gamete.
92. Organisms without cell wall are :
- (1) Eubacteria and slime mould
 - (2) Slime mould and Mycoplasma
 - (3) Mycoplasma and BGA
 - (4) Mycoplasma and *Noctiluca*
93. Which of the following character is not common in algae ?
- (1) Play role in ecology
 - (2) Have pigments
 - (3) Have thalloid body
 - (4) Have terrestrial nature only
94. Which one of the following is not a fungal disease ?
- (1) White Rust of crucifers
 - (2) Loose smut of wheat
 - (3) Red rust of tea
 - (4) Red rot of sugarcane
95. Which of the following description does not apply to phaeophyta ?
- (1) Accessory photosynthetic pigment is fucoxanthin
 - (2) Nearly all are fresh water algae
 - (3) Store food in the form of mannitol
 - (4) Body is filamentous or thalloid
96. If two vegetative fungal cells fuse, It is called :
- (1) Plasmogamy
 - (2) Somatogamy
 - (3) Karyogamy
 - (4) Gametangial copulation
97. Phycomycetes refers to "algae like fungi" as most of the fungi of this group are aquatic. In phycomycetes the mycelium is :
- (1) Multicellular thalloid
 - (2) Coenocytic, septate
 - (3) Unicellular
 - (4) Coenocytic, aseptate
98. In club fungi, which stage is usually prolonged
- (1) Stage between dikaryotisation and karyogamy
 - (2) Stage between karyogamy and meiosis
 - (3) Stage between meiosis and formation of spores
 - (4) Stage between formation of basidia and basidiocarp
99. In mycorrhizal association fungi mainly absorb from the soil.
- (1) P (2) N (3) K (4) Zn
100. Fungi differ from Algae in having
- (1) cell wall made up of cellulose
 - (2) cell wall of chitin and cellulose
 - (3) chlorophyll
 - (4) starch as reserve food
101. Chlorophyll is absent in
- (1) *nostoc* (2) *rhizopus*
 - (3) *funaria* (4) *cycas*
102. Organisms which obtain their food from non-living material in their environment are called
- (1) parasites (2) saprophytes
 - (3) epiphytes (4) sporophytes
103. Facultative saprophyte is
- (1) always parasite
 - (2) parasite but can be saprophyte
 - (3) saprophyte but acquires a parasitic mode
 - (4) always saprophyte
104. Heterothallism was discovered by
- (1) Blakeslee (2) Louis Pasteur
 - (3) Bessey (4) Saccardo

105. Fungal hyphae penetrate hard cell walls of their hosts with the help of
 (1) sugar exudates (2) hormones
 (3) enzymes (4) sharp tips
106. Which of the following sets belongs to the same class of algae?
 (1) *Ectocarpus*, *Ulothrix*, *Porphyra*
 (2) *Chara*, *Polysiphonia*, *Fucus*
 (3) *Sargassum*, *Gracilaria*, *Laminaria*
 (4) *Chlamydomonas*, *Spirogyra*, *Volvox*
107. The fungi which occur on wood are
 (1) epibiotic (2) epicarpic
 (3) epixylic (4) epigeon
108. Select the wrong statements :-
 (1) In Oomycetes, female gamete is smaller and motile, while male gamete is larger and non-motile
 (2) *Chlamydomonas* exhibits both isogamy and an-isogamy and *Fucus* shows oogamy.
 (3) Isogametes are similar in structure, function and behaviour.
 (4) Anisogametes differ either in structure, function or behaviour.
109. The non-motile, greatly thickened asexual spores are called-
 (1) Hypnospores (2) Aplanospores
 (3) Macrozoospores (4) Microzoospores
110. The mycelium is typically coenocytic in
 (1) phycomycetes
 (2) ascomycetes
 (3) basidiomycetes
 (4) none of the above
111. The "perfect stage" of a fungus is when
 (1) it reproduces sexually
 (2) it reproduces asexually
 (3) it forms perfect resting spores
 (4) the fungus is perfectly healthy
112. Sexual reproduction is absent in
 (1) basidiomycetes (2) deuteromycetes
 (3) phycomycetes (4) ascomycetes
113. The female gametangium of ascomycetes is called
 (1) ascogonium (2) cleistothecium
 (3) ascocarp (4) perithecium
114. An open ascocarp is termed
 (1) perithecium (2) cleistothecium
 (3) hysterothecium (4) apothecium
115. Which of the following is achlorophyllous ?
 (1) Algae (2) Bryophytes
 (3) Gymnosperms (4) Slime molds
116. After fusion of sexual gametes in *Mucor* or *Rhizopus*, the resultant structure is known as-
 (1) Oospore (2) Cleistothecium
 (3) Zygosporangium (4) Zygote
117. Haustoria in fungi are meant for
 (1) attachment of mycelium to the host
 (2) increase in the spread of the disease
 (3) reproduction of the fungi
 (4) absorption of nourishment from host cells
118. A plasmodium is
 (1) a uninucleate mass of protoplasm without cell wall
 (2) a mass of protoplasm without nucleus without cell wall
 (3) a multinucleate mass of protoplasm without cell wall
 (4) a multinucleate mass of protoplasm with cell wall
119. The most common constituent of fungal cell walls is
 (1) chitin (2) cellulose
 (3) proteins (4) lipids
120. *Rhizopus* belongs to the class
 (1) ascomycetes
 (2) phycomycetes
 (3) basidiomycetes
 (4) deuteromycetes
121. All life on earth derive its energy directly or indirectly from sun except-
 (1) Mushroom and moulds
 (2) Chemosynthetic bacteria
 (3) Symbiotic bacteria
 (4) Pathogenic bacteria

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| <p>122. Common form of food stored in a fungal cell is-</p> <p>(1) Glycogen (2) Starch
(3) Glucose (4) Sucrose</p> <p>123. Fusion between two morphologically similar but physiologically different thalli in fungi is known as -</p> <p>(1) isogamy (2) anisogamy
(3) homothallism (4) heterothallism</p> <p>124. Algae can be described as</p> <p>(1) autotrophic and chemosynthetic
(2) autotrophic and photosynthetic
(3) heterotrophic and photosynthetic
(4) heterotrophic and chemosynthetic</p> <p>125. Which of the following is a parasitic alga ?</p> <p>(1) <i>Oedogonium</i> (2) <i>Cephaleuros</i>
(3) <i>Sargassum</i> (4) none of these</p> <p>126. A green plant possessing single-celled reproductive organs, no vascular tissue, and no root, stem or leaf should be classified as</p> <p>(1) algae (2) fungi
(3) bryophyte (4) mycoplasma</p> <p>127. All algae have</p> <p>(1) phycobilins and carotene
(2) chlorophyll a and chlorophyll b
(3) chlorophyll b and carotene
(4) chlorophyll a and carotene</p> <p>128. Which one of the following statement is not correct with reference to algae ?</p> <p>(1) all algae are microscopic
(2) algae are thallophytes
(3) four groups of algae are blue-green algae, green algae, brown algae and red algae
(4) algae manufacture food</p> <p>129. Pyrenoids are</p> <p>(1) starch surrounded by protein grains
(2) starch surrounded by oil droplets
(3) protein surrounded by oil droplets
(4) protein surrounded by starch grains</p> <p>130. Red colour of red algae is due to</p> <p>(1) r-phycoerythrin (2) r-phyococyanin
(3) carotenoids (4) xanthophylls</p> | <p>131. Floridean starch as a stored photosynthate is present in</p> <p>(1) red algae (2) green algae
(3) blue-green algae (4) brown algae</p> <p>132. Storage produced in Rhodophyceae is</p> <p>(1) starch and oil (2) cellulose
(3) floridean starch (4) glycogen</p> <p>133. Female sex organ of Rhodophyceae is termed</p> <p>(1) carpogonium (2) oogonium
(3) sporogonium (4) archegonium</p> <p>134. Heterothallism refers to-</p> <p>(1) Fusion is not accompanied with zygote formation
(2) Fusion between morphologically similar strain
(3) Fusion between the strains of structurally similar and physiologically different
(4) All the above</p> <p>135. DNA is transcribed by some viral RNA using the enzyme</p> <p>(1) DNA polymerase
(2) Reverse transcriptase
(3) Endonuclease
(4) Ligase</p> <p>136. During protein synthesis the correct amino acid is brought to the ribosome and inserted in correct sequence in the growing polypeptide chain by</p> <p>(1) base pairing between DNA and mRNA
(2) base pairing between mRNA codons and anticodon in tRNA
(3) interaction between tow anticodons
(4) interaction between two triplet codons</p> <p>137. Semiconservative mode of DNA replication was reported by</p> <p>(1) Watson and Crick
(2) Messelson and Stahl
(3) Kornberg and Ochoa
(4) Luria and Delbruck</p> |
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138. Khorana got the Nobel Prize on –
 (1) gene synthesis
 (2) genetic code
 (3) protein synthesis
 (4) enzyme synthesis
139. Which one of the following pairs is correctly matched –
 (1) ribosomal RNA Carries amino acids to the site of protein synthesis
 (2) transcription process by which protein is carries the information from nucleus to ribosomes
 (3) translation process by which mRNA carries the information from the nucleus to ribosomes
 (4) anticodon site of a tRNA molecule that binds to the m-RNA molecule
140. Reverse transcriptase is
 (1) RNA dependent RNA polymerase
 (2) DNA dependent RNA polymerase
 (3) DNA dependent DNA polymerase
 (4) RNA dependent DNA polymerase
141. Which one of the following chemical characteristics is not common to all living beings
 (1) similar triplet codes for amino acids
 (2) energy is stored by high energy-rich phosphate bonds
 (3) types of protein present in the body
 (4) ribosomes act as sites for protein synthesis
142. Initiation of polypeptide chain in protein synthesis is induced by
 (1) methionine (2) glycine
 (3) leucine (4) lysine
143. Khorana synthesized a biologically functional tyrosine suppressor t-RNA of *E. coli* in 1979 it contained.
 (1) 77 nucleotide pairs
 (2) 207 nucleotide pairs
 (3) 312 nucleotide pairs
 (4) 333 nucleotides only
144. The function of a non-sense codon is
 (1) to release polypeptide chain from t-RNA
 (2) to form an unspecified amino acid
 (3) to terminate the message of gene controlled protein synthesis
 (4) to convert a sense DNA into non-sense DNA
145. From a DNA template with the sequence CTGATAGC, the mRNA sequence formed would be
 (1) GUCTUTCG
 (2) GACUAUCG
 (3) UACTATCU
 (4) GAUTATUG
146. Which is the correct sequence of code transfer involved in the formation of a polypeptide ?
 (1) DNA, tRNA, tRNA, mRNA
 (2) rRNA, DNA, mRNA, tRNA
 (3) mRNA, tRNA, DNA, amino acids
 (4) DNA, mRNA, tRNA, amino acids
147. Genetic code consists of
 (1) adenine and guanine
 (2) guanine and cytosine
 (3) cytosine and uracil
 (4) all
148. The smallest unit of DNA which is capable of undergoing crossing over and recombination is called
 (1) recon (2) muton
 (3) cistron (4) intron
149. The modern concept of gene is that it is
 (1) a segment of DNA capable of crossing over
 (2) a functional unit of DNA
 (3) a segment of DNA
 (4) a segment of chromosome

150. Indicate the order in which the following steps occur in the production of a mature mRNA
- (1) initiation of transcription, splicing, addition of 5' cap, addition of poly A tail, transport to cytoplasm
 - (2) initiation of transcription, addition of 5' cap, splicing, addition of poly A tail, transport to cytoplasm.
 - (3) initiation of transcription, addition of poly A tail, addition of 5' cap, splicing, transport to cytoplasm
 - (4) initiation of transcription, addition of 5' cap, addition of poly A tail, splicing, transport to cytoplasm
151. Which DNA molecule is more stable if the nucleotide pairs are 80
- (1) 40 A = T & 40 C = G
 - (2) 30 A = T & 50 C = G
 - (3) 60 A = T & 20 C = G
 - (4) 20 A = T & 60 C = G
152. During transcription, the DNA site at which RNA polymerase binds is called
- (1) promoter
 - (2) receptor
 - (3) enhancer
 - (4) regulator
153. During translation, the mRNA site at which Ribosome binds is called
- (1) promoter
 - (2) shine-dalgarno sequence
 - (3) regulator
 - (4) tata box
154. A Transcription factor that recognizes bacterial promoter sequences and facilitates the binding of RNA polymerase to the promoter is called
- (1) transcription factor P
 - (2) sigma factor
 - (3) transcription factor K
 - (4) ribozyme
155. Which of the following Amino acid is encoded by the codon UGA in mammalian mitochondria
- (1) tryptophan
 - (2) glycine
 - (3) stop codon
 - (4) not any amino acid
156. rRNA except 5 s rRNA in eukaryotes is synthesized by
- (1) RNA polymerase I
 - (2) RNA polymerase II
 - (3) RNA polymerase III
 - (4) DNA polymerase
157. In DNA Replication RNA primer is synthesized by
- (1) DNA polymerase I
 - (2) DNA polymerase III
 - (3) primosome
 - (4) primase
158. The coding sequences of mRNA is
- (1) intron
 - (2) exon
 - (3) muton
 - (4) cistron
159. The removal of non-coding sequence from primary eukaryotic RNA sequence is called
- (1) splicing
 - (2) post transcriptional modification
 - (3) intranuclear traffic control signal
 - (4) none of the above
160. DNA replication is
- (1) continuous and conservative
 - (2) discontinuous and semiconservative
 - (3) Semidiscontinuous and semiconservative
 - (4) Conservative and semidiscontinuous
161. What becomes established from the wobble hypothesis ?
- (1) Process of peptide chain elongation
 - (2) Economy of the number of t-RNA molecules
 - (3) Process of peptide chain initiation
 - (4) Process of peptide chain termination
162. DNA generally acts as a template for the synthesis of
- (1) only protein
 - (2) only DNA
 - (3) only RNA
 - (4) both DNA and RNA

163. Histone protein are :-

- (1) Basic, negatively charged
- (2) Basic, positively charged
- (3) acidic, positively charged
- (4) acidic, negatively charged

164. Which one is correctly matched :-

- (1) SnRNA → Spliceosome
- (2) RNA polymerase → Splicing
- (3) Okazaki fragment → transcription
- (4) Cistron → Non coding part

165. If a linear double stranded DNA molecule has 2000 bps then how many phosphodiester bonds and glycosidic bonds would be presents in it respectively ?

- (1) 2000 & 2000 (2) 3998 & 4000
- (3) 4000 & 4000 (4) 4000 & 8000

166. Which of the following statements is correct about DNA polymerase ?

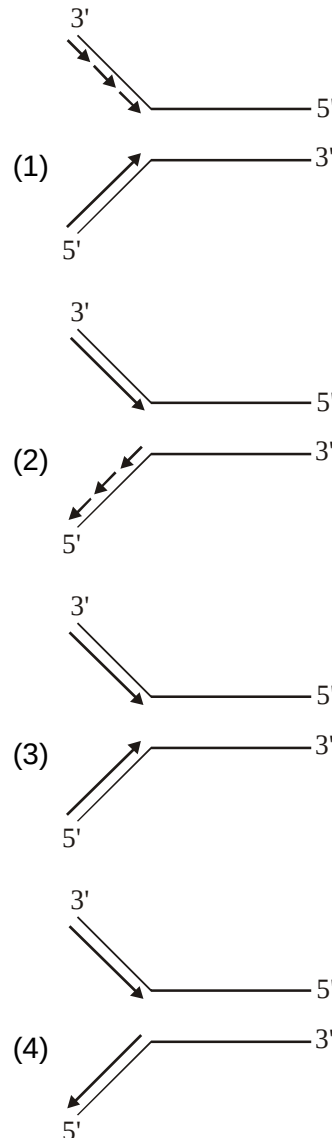
- (1) DNA polymerase can synthesize mRNA in the 3' to 5' direction
- (2) DNA polymerase can synthesize DNA in the 5' to 3' direction
- (3) DNA polymerase can synthesize mRNA in the 5' to 3' direction
- (4) DNA polymerase can synthesise DNA in the 3' to 5' direction

167. Match the enzymes [given in column I] with their function given in column II and choose the correct combination from the given option :-

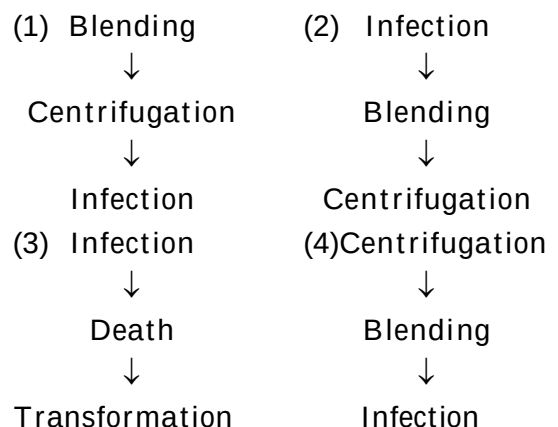
Column I		Column II	
(A)	Helicase	(I)	Joining of nucleotides
(B)	Gyrase	(II)	Opening of DNA
(C)	Primase	(III)	Release the tension of super coiling
(D)	DNA polymerase III	(IV)	RNA primer

- (1) A-II, B-I, C-III, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-IV, D-III
- (4) A-II, B-III, C-I, D-IV

168. Which one is true among the following for DNA replication :-



169. The correct sequence of steps in Hershey and Chase's experiment that provided unequivocal proof that DNA is the genetic material was :-



170. Observe the following carefully :-

- Formation of primer
 - 5'→ 3' polymerization
 - use of sigma(σ) factor
 - Requirement of ori
 - Phosphodiester bond formation
 - Both strands of DNA used as template
 - Requirement of promoter
 - Discontinuous polymerization
- Find the correct match

	Common to replication and transcription	Only in replication	Only in transcription
1	b,e,c	a,d,f	h,g
2	b,e	a,d,f,h	c,g
3	a,f,h	b,c,g	d,e
4	b,e	d,f,h	a,c,g

171. Bonding between deoxyribose and base in purine nucleoside molecule is :-

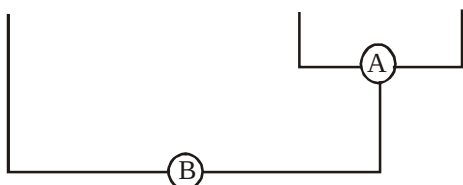
- 1'1' glycosidic linkage
- 1'-6' glycosidic linkage
- 1'-9 glycosidic linkage
- 1'-4' glycosidic linkage

172. DNA & RNA differ from each other with respect to :-

- Nitrogen bases only
- Sugar and phosphate
- Nitrogen bases and phosphate
- Nitrogen bases and sugar

173. What does A and B represent in the given representation ?

Phosphoric acid + Sugar + Nitrogen Base



- A- Nucleoside, B- Nucleotide
- A- Nucleotide, B- Nucleoside
- A- Nucleic acid, B- Nucleotide
- A- Nucleoside, B- Ribonucleoside

174. RNA can act as genetic material in:

- All organisms
- All viruses
- Some viruses
- All bacteria

175. Select the correct statement:

- DNA polymerase has high degree of accuracy
- During DNA replication, only one strand of DNA act as template for synthesis of new strand
- Taylor has proved by his experiment that DNA is the genetic material
- E. coli has 4.6×10^9 bp in its genetic material

176. In his transforming experiments on mice, Griffith has taken 2 strains of streptococcus pneumonia: R-strain and S-strain. in which of the following case mice will remain alive?

- Heat killed S-strain + R-strain + Proteases injected into mice
- Heat killed S-strain + R-strain + RNases injected into mice
- Heat killed S-strain + R-strain + DNases injected into mice
- DNA from heat killed S-strain + R-strain injected into mice

177. Match the columns:

Column A		Column B	
A.	Taylor and colleagues	(i)	Biochemical characterization of transforming principle
B.	Meischer	(ii)	Nuclein
C.	Crick	(iii)	Central dogma
D.	Avery, Macleod and McCarthy	(iv)	Semi-conservative replication of DNA

- A-(iv), B-(ii), C-(iii), D-(i)
- A-(iv), B-(ii), C-(i), D-(iii)
- A-(ii), B-(iv), C-(i), D-(iii)
- A-(iv), B-(i), C-(iii), D-(ii)

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| <p>178. In prokaryotes like E.coli 1.... charged DNA is present in large loops held by2..... charged3..... to form nucleoid.</p> <p>(1) 1-Positively, 2-Negatively, 3-Fatty acids</p> <p>(2) 1-Negatively, 2-Positively, 3-Proteins</p> <p>(3) 1-Negatively, 2-Positively, 3-Fatty acids</p> <p>(4) 1-Positively, 2-Negatively, 3-Proteins</p> <p>179. Select the incorrect statement about DNA</p> <p>(1) DNA is an acidic substance</p> <p>(2) DNA act as genetic material in most of organisms</p> <p>(3) In bacteriophage λ, 48502 base pairs are found in DNA</p> <p>(4) F. Meischer first identified the DNA in 1829</p> | <p>180. In symbiotic association between algae and fungi which one is not correct about lichen:</p> <p>(1) Algae prepare food for fungi and itself also</p> <p>(2) Fungi provide shelter and absorb mineral and water for its partner</p> <p>(3) Lichens are very good pollution indicator because they grow in polluted areas only</p> <p>(4) In lichens both partners are so close in their association that one would neve imagine that they had two different organisms within them</p> |
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Space For Rough Work
